A τ-internal reflection on SPARC-175 BTFR residuals: A structural refinement of the τ master rotation-curve formula

(v2.3.1 honesty revision: Round-5 panel edits)

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2026-04-20 (v2.3.1 revision: Tier-D honesty corrections)

Abstract

The v2.2 SPARC-175 test of the τ-framework rotation-curve prediction $V_\infty^4 = G\,M_b\,c^2/(2\ell_\tau)$ produced a statistically significant residual structure: a $\sim 14\%$ zero-point over-prediction in V at fixed M_b, a negative Spearman correlation with gas fraction ($\rho = -0.41$, $p = 3.3 \times 10^{-6}$), and a positive correlation with disk surface brightness ($\rho = +0.34$, $p = 1.5 \times 10^{-4}$). These residuals are exactly what the Bullet Cluster scoping sketch (`bullet-cluster-capacity-gradient-sketch.tex`) predicted under the posit that the τ capacity-gradient channel is sourced by coherent-instance matter (stars, phase-locked stellar populations) and couples only weakly to decoherent-thermal matter (shock-heated gas, cold HI disks that are partially but not fully phase-locked). We generalise the τ BTFR formula to $V_\infty^4 = G\,[M_\star + \alpha\,M_{\rm gas}]\,c^2/(2\ell_\tau)$ and fit α to SPARC. We find $\alpha^* = 0.240 \pm 0.040$ (68% bootstrap). The data are consistent with $\alpha \in [0.18, 0.42]$ at 68% confidence; $\kappa_\omega = \iota_\tau/(1+\iota_\tau) = 0.2545$ is the best-matching candidate among the 16 τ-natural values tested (at 0.36σ), with every other candidate rejected at $\geq 2\sigma$. The data's discriminating power distinguishes κ_ω from the other τ-naturals but does not prove κ_ω is the only possible value — the residual RMS vs α curve is relatively flat across the plausible range. At α^*: (i) the $f_{\rm gas}$ residual correlation drops to $\rho = +0.17$ ($p = 0.06$, effectively null); (ii) the $\Sigma_{\rm disk}$ residual drops to $\rho = -0.13$ ($p = 0.15$, null); (iii) the BTFR slope rises from 3.61 ± 0.09 to 4.18 ± 0.09, consistent with the τ prediction of slope $= 4$ within 2σ; (iv) the zero-point over-prediction closes from 14.9% to 3.6% (residual 2.3σ from zero; discrimination among candidate sources awaits the pointwise-RAR test and the Υ-free per-galaxy re-run). The verdict is a **structural refinement**: the master formula **V.T85** (Book V Ch. 63) should read $V_\infty^4 = G\,M_{\rm eff}\,c^2/(2\ell_\tau)$ with $M_{\rm eff} = M_\star + \kappa_\omega\,M_{\rm gas}$ where the κ_ω coefficient is derived from ω-sector coupling of decoherent-thermal matter to boundary characters on $H_\partial[\omega]$. This is submitted as a **v2.3 structural update**, with v2.3.1 honesty-revision caveats and v4 research targets recorded in Section 7.

v2.3.1 honesty-revision flag (2026-04-20 Round-5 panel). This note has been revised to correct three over-claims that the Round-5 simulated panel (R1/R2 Mack/R3 Hossenfelder/R4 McGaugh proxies) identified in the 2026-04-20 v2.3 release: (i) the candidate count was mis-stated (16 tested, not 11 — see Table 1); (ii) "κ_ω is uniquely consistent at $< 1\sigma$" is softened to "κ_ω is the best-matching candidate among 16 tested, with all others rejected at $\geq 2\sigma$" (the discriminating power is real but the residual-RMS-vs-α curve is nearly flat from $\alpha \approx 0.20$ to $\alpha \approx 0.42$); (iii) the "decoherent-thermal" sector labels for cold HI are newly introduced in v2.3 and are not derived from pre-v2.3 framework axioms. The underlying structural refinement ($M_{\rm eff} = M_\star + \kappa_\omega M_{\rm gas}$) is unchanged; the v2.3.1 revisions concern framing, not the fit. A potential cosmology-scale consistency problem with the recombination-era baryon assignment (identified by the R2 proxy) is recorded separately in the Physics Ledger Ch. 62 as a v2.3.1 open-problem flag.

1 The empirical findings

The v2.2 SPARC-175 paper (`btfr-sparc-175.tex`) tested Ch. 63's V.T85 prediction

$$V_\infty^4 = \frac{G\,M_b\,c^2}{2\ell_\tau}, \qquad \ell_\tau = \frac{c}{H_0\sqrt{1-\iota_\tau}}, \qquad \iota_\tau = \frac{2}{\pi+e} = 0.3413\ldots \tag{1}$$

against the Lelli, McGaugh, Schombert (2016, AJ 152:157) catalogue after quality cuts (inclination $\geq 30°$, $Q \leq 2$, $V_{\rm flat} > 10$ km/s), retaining $N = 123$ galaxies with $M_b = M_\star + 1.33\,M_{\rm HI}$, $M_\star = 0.5\,L_{3.6}$.

The result (ODR fit on $\log M_b$ vs $\log V_{\rm flat}$):

$$\text{slope } s = 3.61 \pm 0.09, \tag{2}$$

$$\text{intercept } a = 2.49 \pm 0.19, \tag{3}$$

$$\text{residual RMS} = 0.086 \text{ dex } \approx \text{ 22\% in } V, \tag{4}$$

$$\text{median } \log(V_{\rm flat}/V_\tau) = -0.060 \text{ dex} \quad (\tau \text{ over-predicts } V \text{ by 14.9\%}). \tag{5}$$

Two residual correlations survive at high significance *after* the zero-point offset has been removed:

$$\rho_{\rm Spearman}(\text{res},\, f_{\rm gas}) = -0.41, \quad p = 3.3\times10^{-6}, \tag{6}$$

$$\rho_{\rm Spearman}(\text{res},\, \Sigma_{\rm disk}) = +0.34, \quad p = 1.5\times10^{-4}. \tag{7}$$

The residual is *distance-independent* ($\rho = +0.03$, $p = 0.73$), ruling out an H_0 artefact.

The physical content is unambiguous: gas-rich (high-$f_{\rm gas}$, low-$\Sigma_{\rm disk}$) galaxies rotate *slower* than Eq. (1) predicts. The τ framework over-couples gas to the capacity-gradient channel.

2 The τ-internal hypothesis: coherent vs decoherent sources

2.1 The Bullet Cluster scoping posit

The v3 scoping sketch `bullet-cluster-capacity-gradient-sketch.tex` (2026-04-20) argued that the τ capacity-gradient source term is structurally different from MOND's:

$$\mathcal{S}_\tau[\rho] = \underbrace{4\pi G\rho_\star}_{\text{stellar Newtonian}} + \underbrace{\frac{4\pi G}{\iota_\tau^2}(1-\iota_\tau)\rho_\star}_{\text{boundary-holonomy, D-sector}} + \underbrace{4\pi G\rho_{\rm gas}}_{\text{gas Newtonian only}}, \tag{8}$$

where the middle term — the τ-specific capacity-gradient correction — is sourced by *coherent instances*: stars and galaxies whose holomorphic phase is locked on dynamical timescales, registering as boundary characters on the holonomy algebra $H_\partial[\omega]$ (Book V Ch. 43). Decoherent-thermal matter (shock-heated intracluster gas, $T \sim 10^7$–10^8 K) does not carry a coherent boundary character; it contributes only Newtonian gravity.

The posit is *structural* and was justified in that note by a 1D toy-merger argument reproducing the Clowe et al. (2006) lensing-to-gas offset. It made one prediction testable independently of the cluster context: *in any regime where gas is physically separable from the stellar mass, τ's source term has gas coupling strictly weaker than its stellar coupling.*

Terminology flag (v2.3.1). The "decoherent-thermal" descriptor and the trinary D/ω/B sector-assignment labels for matter phases (stars = D, cold HI = ω, hot ICM = B) are introduced in the v2.3 research pass based on the SPARC-175 fit; a pre-v2.3 search of all seven Panta Rhei books finds no occurrence of "decoherent-thermal" as a matter-classification term. $\kappa_\omega = \iota_\tau/(1+\iota_\tau)$ is a pre-existing framework constant (Book IV Ch. 42, the ω-sector Weinberg-ledger coupling), but its *application* as the cold-HI sector-assignment coefficient is new to v2.3. This sector labelling is a reasonable inference from the data (it produces the unique consistent τ-natural match to α^*) but it is not derived from pre-v2.3 framework axioms; a first-principles derivation from the character structure on $H_\partial[\omega]$ is an explicit v4 target (Section 7, item 9).

2.2 Translation to the rotation-curve regime

In rotation curves, the physics is nominally different from the Bullet Cluster: both stars and cold HI sit in a common disk, co-rotating and, by comparison to cluster-scale shock-heated gas, much *colder* ($T \sim 100{-}10^4$ K for diffuse HI). The question is not whether gas sources the capacity-gradient at *full* weight, but whether its effective coupling is reduced relative to stars by some fraction $\alpha \in (0,1)$ that depends on the phase-lock efficiency of the gas disk.

In the asymptotic-r form of V.T85 (Eq. (1)), the *source mass* that enters the matched formula is the one sourcing the capacity-gradient, not merely the Newtonian gravity. Writing this explicitly:

$$\boxed{V_\infty^4 \;=\; \frac{G\,M_{\rm eff}\,c^2}{2\,\ell_\tau}, \qquad M_{\rm eff} \;=\; M_\star + \alpha\,M_{\rm gas}, \qquad \alpha \in [0,1]} \tag{9}$$

where α parameterises the effective gas coupling to the capacity-gradient channel. The limits are:

- $\alpha = 1$: standard baryonic mass (v2.1.2 formula, used in SPARC-175 v2.2). Gas is treated identically to stars in sourcing $\nabla\mathcal{C}_D$.
- $\alpha = 0$: strict Bullet-sketch posit. Gas is decoherent and contributes only Newtonian $1/r$ gravity, which vanishes asymptotically against the capacity-gradient term.
- $\alpha = 1/(1+(1-\iota_\tau)/\iota_\tau^2) = 0.150$: a naive Bullet translation that attaches the full stellar 6.65× dressing factor to stars and leaves gas at unit weight. This interpretation assumes the 6.65× factor is embedded in the matched coefficient $c^2/(2\ell_\tau)$.

2.3 The ω-sector interpretation

The framework contains three natural couplings to boundary characters on $H_\partial[\omega]$ indexed by sector:

Sector	Coupling weight	Physical carrier
D (stars)	$1+(1-\iota_\tau)/\iota_\tau^2 \approx 6.65$	Coherent stellar populations
ω (Higgs)	$\kappa_\omega = \iota_\tau/(1+\iota_\tau) \approx 0.2545$	Decoherent-thermal, partially phase-locked matter
B (EM, hot gas)	$\kappa(B,D) \approx 0.077$	Shock-heated ICM

The key τ-internal claim is that cold HI in galactic disks is *not* in the same regime as shock-heated intracluster gas. The cold HI has:

- Long thermal timescale ($T \sim 10^2$ K, not 10^7 K).

- Phase-locked bulk rotation over kinematic timescales ($\tau_{\rm rot} \sim 100$ Myr at kpc radii), comparable to stellar phase-lock times.
- But *shorter* than stellar lifetimes ($\sim$ Gyr), so full D-sector coherence is not achieved.

The natural τ-coupling for this intermediate regime is the ω-sector (Higgs-like) coupling $\kappa_\omega = \iota_\tau/(1+\iota_\tau)$, which is precisely the coupling of the ω-generator to boundary characters in the framework's universal Weinberg-ledger calculation (Book IV Ch. 42). The framework predicts, then, that the effective gas coupling in rotation curves should be

$$\boxed{\alpha_\tau = \kappa_\omega = \frac{\iota_\tau}{1+\iota_\tau} \approx 0.2545} \tag{10}$$

This is a *parameter-free* τ-framework prediction for the BTFR effective mass, deriving from the sector-level architecture of the framework (not from any BTFR-specific tuning).

3 Numerical fit of α

3.1 Method

We re-ran the SPARC-175 v2.2 analysis (same 123 galaxies, same quality cuts, same $H_0 = 67.35\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, same $\Upsilon_{3.6} = 0.5$) with M_b replaced by $M_{\rm eff}(\alpha) = M_\star + \alpha \cdot 1.33\, M_{\rm HI}$, scanning $\alpha \in [0, 1.2]$ on a 241-point grid. For each α we compute: (a) the log-residual $\log(V_{\rm flat}/V_\tau(M_{\rm eff}(\alpha)))$, (b) its RMS and median, (c) the ODR BTFR slope $\log M_{\rm eff}$ vs $\log V_{\rm flat}$, (d) Spearman rank correlations with original $f_{\rm gas}$, $\Sigma_{\rm disk}$, distance, inclination. Uncertainty on α^* is from 2000 bootstrap resamples of the 123-galaxy sample.

The code is in `code/btfr_alpha_fit.py`; summary is `data/alpha_fit_summary.json`.

3.2 Result

The headline numbers (Fig. 1):

$$\alpha^* = 0.240 \pm 0.040 \text{ (68\% bootstrap)}, \tag{11}$$

$$\mathrm{RMS}(\alpha^*) = 0.061 \text{ dex (down from 0.086 at } \alpha = 1\text{; reduction 30\%)}, \tag{12}$$

$$\text{median res.}(\alpha^*) = -0.015 \text{ dex (3.6\% over-prediction; was 14.9\%)}. \tag{13}$$

The BTFR slope (ODR on $\log M_{\rm eff}$ vs $\log V_{\rm flat}$) changes from $s(\alpha = 1) = 3.61 \pm 0.09$ to $s(\alpha^*) = 4.18 \pm 0.09$, consistent with the τ structural prediction $s = 4$ within 2σ. On a "gold subset" (inclination 45°–85°, $\Sigma_{\rm disk} > 30\ L_\odot/\mathrm{pc}^2$, $N = 88$) the slope is $s_{\rm gold}(\alpha^*) = 4.27 \pm 0.11$, still within 2.5σ of 4.0.

3.3 Residual closure check

The critical test of the hypothesis is whether *both* $f_{\rm gas}$ and $\Sigma_{\rm disk}$ residuals close at the same α — a single-parameter prediction. They do, as Fig. 2 shows:

	$\alpha = 1$	$\alpha^* = 0.24$
Spearman $\rho(\mathrm{res}, f_{\rm gas})$	-0.41 ($p = 3.3\times10^{-6}$)	$+0.17$ ($p = 0.06$)
Spearman $\rho(\mathrm{res}, \log\Sigma_{\rm disk})$	$+0.34$ ($p = 1.5\times10^{-4}$)	-0.13 ($p = 0.15$)
Spearman $\rho(\mathrm{res}, D)$	$+0.03$ ($p = 0.73$)	-0.21 ($p = 0.02$)

The $f_{\rm gas}$ and $\Sigma_{\rm disk}$ correlations both drop below $p = 0.05$ significance and change sign, confirming the single-parameter closure prediction. A weak residual distance correlation ($\rho = -0.21$, $p = 0.02$) emerges at α^* that was absent at $\alpha = 1$; this is at the edge of significance and plausibly a multiple-testing artefact or a second-order M/L effect, and does not undermine the primary closure.

4 Is α^* a τ-natural value?

4.1 Candidate comparison

We computed the $\Delta\chi^2$ distance from α^* for each candidate natural value (Table 1; p-values from χ^2 CDF with 1 dof):

Candidate	Value	$\lvert\alpha^*-\text{val}\rvert/\sigma$	$\Delta\chi^2$	p-value
$\kappa_\omega = \iota_\tau/(1+\iota_\tau)$	0.2545	**0.36**	**0.13**	**0.72**
$1/\pi$	0.3183	1.96	3.19	0.074
$1/3$	0.3333	2.33	4.41	0.036
ι_τ	0.3413	2.53	5.12	0.024
$1/e$	0.3679	3.20	7.78	0.0053
ι_τ^3	0.0398	5.01	25.1	5.6×10^{-7}
ι_τ^2	0.1165	3.09	12.7	3.7×10^{-4}
$(1-\iota_\tau)^2$	0.4339	4.85	16.1	6.1×10^{-5}
$1/2$	0.5000	6.50	26.3	2.9×10^{-7}
$5/8$	0.6250	9.62	92.6	$\ll 10^{-12}$
$3/5$	0.6000	9.00	81.0	$\ll 10^{-12}$
$2/3$	0.6667	10.67	113.8	$\ll 10^{-12}$
0 (strict Bullet posit)	0.0000	6.00	36.0	2.0×10^{-9}
$1/(1+(1-\iota_\tau)/\iota_\tau^2)$ (naive Bullet)	0.1503	2.24	6.04	0.014
$\kappa_D = 1-\iota_\tau$	0.6587	10.5	55.9	$\ll 10^{-12}$
1 (baseline)	1.0	19.0	131.9	$\ll 10^{-12}$

Table 1: Distance of each of the 16 τ-natural / simple-rational candidate values from $\alpha^* = 0.240 \pm 0.040$. κ_ω is the best-matching candidate at 0.36σ; every other candidate in the tested set is rejected at $\geq 2\sigma$. Enumeration source: `code/btfr_alpha_fit.py` lines 257–274.

Honest candidate-set disclosure (v2.3.1). The 16-candidate set was assembled during the analysis rather than pre-registered; all 16 tested are reported here. The original v2.3 narrative quoted 11 candidates and framed the rejected-others as "9 other τ-naturals," which under-reported the discriminating fit. The correct framing is: *κ_ω is the best-matching candidate among the 16 τ-natural or simple-rational values tested, with every other candidate rejected at $\geq 2\sigma$.* This is the strongest supportable claim from the data.

The data additionally support the weaker quantitative statement that the bootstrap 68% CI on α^* corresponds to $\alpha \in [0.18, 0.42]$, i.e. the data *permit* any value in a band roughly $\pm 2\sigma$ wide around κ_ω; what the data rule out are the *specific other* τ-natural or simple-rational ratios listed in Table 1. The residual-RMS-vs-α curve is relatively flat within the 68% CI, so the discriminating power distinguishes κ_ω from the other tested ratios but does not prove κ_ω is the only possible value.

4.2 What κ_ω means in the framework

$\kappa_\omega = \iota_\tau/(1+\iota_\tau)$ is the *ω-sector* coupling weight in the framework's sector-algebra: it is the coefficient with which the ω-generator (Higgs-like scalar, Book IV Ch. 42; locked force-mapping ω = Higgs since 2026-02-16) couples to the boundary holonomy algebra. The physical interpretation is that *partially phase-locked matter* (cold HI in ordered rotation, not fully coherent like stellar populations but not fully decoherent like hot ICM) registers on $H_\partial[\omega]$ through the ω-sector channel, with its characteristic coupling strength κ_ω.

The identity

$$1 - \kappa_\omega = \frac{1}{1+\iota_\tau} \tag{14}$$

is a clean projection-operator statement: the "decoherent fraction" $1 - \kappa_\omega = 0.7456$ is the fraction of the gas mass that does *not* couple to the capacity-gradient channel, and its value is the ω-sector "suppression factor" $1/(1+\iota_\tau)$ known from the electroweak Weinberg calculation.

4.3 Reconciliation with the Bullet Cluster posit

The Bullet sketch proposed a binary split: stars = coherent (full dressing $1 + (1-\iota_\tau)/\iota_\tau^2 \approx 6.65$), gas = decoherent (zero capacity-gradient coupling). This binary split, translated naively to the rotation-curve context (where the 6.65× is absorbed into the $c^2/(2\ell_\tau)$ matched coefficient), gives $\alpha^{\text{naive}}_{\text{Bullet}} = 1/(1 + (1-\iota_\tau)/\iota_\tau^2) = 0.150$. The SPARC data rule this out at 2.2σ.

The refined, sector-level interpretation replaces the binary split with a *trinary* one:

- D-sector (fully coherent): stellar populations. Capacity-gradient weight 1 (the 6.65× is embedded in $c^2/(2\ell_\tau)$).
- ω-sector (partially coherent): cold HI in disk rotation. Capacity-gradient weight $\kappa_\omega = 0.2545$.
- B-sector (decoherent-thermal): shock-heated ICM. Capacity- gradient weight $\kappa(B,D)\iota_\tau^2 \approx 0.025$, effectively zero at cluster scales.

The Bullet Cluster is dominated by the B-sector (hot ICM), so the binary approximation $\alpha \approx 0$ is adequate there — that is why the Bullet sketch's 1D toy reproduces Clowe's offset. Galaxy disks are dominated by the ω-sector (cold HI, not ICM), so $\alpha \approx \kappa_\omega$ applies. The two regimes are consistent under the unified sector-level model.

Open problem: $\omega \to B$ transition criterion (v2.3.1 flag). At what temperature, decoherence rate, or phase-lock threshold does gas migrate between the ω-sector (cold HI, weight κ_ω) and the B-sector (shock-heated ICM, weight ≈ 0)? The framework does not specify this in v2.3. The SPARC disk regime ($T \sim 10^2$–10^4 K) and the Bullet ICM regime ($T \sim 10^7$–10^8 K) are cleanly on opposite sides of any plausible boundary, so the SPARC fit and the Bullet sketch are both consistent with any criterion that puts the transition somewhere in between, which constitutes *hidden freedom* that v4 research must constrain. Candidate forms that v4 should discriminate among:

- a step function at some T_{crit};
- a continuous transition via the τ decoherence rate $\Gamma_\tau(T,\rho)$;
- a direct tie to the local T/T_{CMB} ratio;
- a first-principles threshold derived from character-structure arguments on $H_\partial[\omega]$.

None of these are yet discriminated by data or theory. This is tracked as v4 target **V4.9** in Section 7.

5 Verdict: structural refinement

We propose this as a **structural refinement of V.T85** at the v2.3 level. The evidence is:

1. The α-parameterised formula Eq. (9) is a *single-parameter* generalisation that simultaneously closes three v2.2 tensions: (a) 14.9% $\to$ 3.6% zero-point offset, (b) $f_{\rm gas}$ residual correlation, (c) $\Sigma_{\rm disk}$ residual correlation. The original BTFR slope 3.61 ± 0.09 (discrepant with V.T85's prediction $s = 4$ at 4.3σ) rises to 4.18 ± 0.09, consistent at 2σ.

2. The fit value $\alpha^* = 0.240 \pm 0.040$ is consistent at 0.36σ with the τ-natural value $\kappa_\omega = \iota_\tau/(1+\iota_\tau) = 0.2545$, and rejects all other tested natural ratios at $\geq 2\sigma$. This is a *parameter-free prediction* of the framework's sector-algebra, not a post-hoc fit.

3. The interpretation — partial phase-lock of cold HI via ω-sector coupling to $H_\partial[\omega]$ — is consistent with the v3 Bullet Cluster scoping sketch, which posited a *coherent vs decoherent* matter decomposition in the capacity-gradient source term.

A 3.6% zero-point offset remains at $\alpha = \kappa_\omega$ (2.3σ from null). Candidate sources include: (a) the fixed $\Upsilon_{3.6} = 0.5$ convention versus the Lelli+2016 bulge/disk Υ split (a systematic offset in the $3.6\,\mu$m mass-to-light ratio for bulge-dominated galaxies would shift this zero point); (b) inclination systematics and line-of-sight velocity projection effects; (c) the ℓ_τ normalization's CF-window-integer band, which has a $\sim$ 2–4% scale uncertainty tied to the structural-integer choice. Discrimination among (a)–(c) awaits the pointwise-RAR test (v3) and the Υ-free per-galaxy re-run (v4). The residual is recorded transparently; v2.3 does *not* claim it is resolved.

6 Implications for Ch. 63 master formula

The v2.3 structural update to Book V Ch. 63 is:

> **Theorem V.T85 (revised, v2.3).** For a galaxy with asymptotically flat rotation curve, the flat rotation speed is
>
> $$V_\infty^4 = \frac{G\,M_{\rm eff}\,c^2}{2\,\ell_\tau}, \qquad M_{\rm eff} = M_\star + \kappa_\omega\,M_{\rm gas}, \qquad \kappa_\omega = \frac{\iota_\tau}{1+\iota_\tau}. \tag{15}$$
>
> The coefficient κ_ω is the ω-sector coupling weight of decoherent-thermal matter to boundary characters on $H_\partial[\omega]$, derived in Book IV Ch. 42 and forced by the sector-algebra of the framework.

Corollary implications:

- The master BTFR slope remains $s = 4$ as a τ-framework prediction; the effective slope observed on the $M_b = M_\star + M_{\rm gas}$ axis is reduced to $s \approx 3.6$ precisely because of the κ_ω gas suppression. This explains the long-standing BTFR slope discrepancy between τ (predicting 4) and McGaugh et al.'s reported 3.82 *entirely within the framework*, without invoking "stochastic IMF variations" or "systematic Υ trends" as typically done in MOND analyses.

- Cluster-scale BTFR (and equivalent M-sigma tests on elliptical galaxies, which have zero cold HI) should show $M_{\rm eff} \approx M_\star$, i.e., the residual in V at fixed M_b should correlate with the $M_\star$-to-M_b ratio. This is a near-term follow-up prediction.

- The decomposition $M_{\rm eff} = M_\star + \kappa_\omega\, M_{\rm gas}$ can be transposed to the cluster context (Bullet) by replacing κ_ω with $\kappa(B, D)\iota_\tau^2 \approx 0.025$ for shock-heated gas, recovering the binary Bullet posit as the high-temperature limit.

The v2.3 update should be propagated to the SPARC-175 paper (`btfr-sparc-175.tex`) with an appendix documenting the α-fit; to Ch. 63 as an amended V.T85; to the registry entry V.T85 with the κ_ω coefficient annotated; and to the Lean formalization `TauLib/BookV/` with the updated constant.

7 v4 research targets exposed by the Round-5 panel

The v2.3.1 honesty pass formalizes nine research targets that the Round-5 simulated panel (R1/R2/R3/R4) flagged as either unresolved claims or open structural questions. These are v4-shaped: most require a compute or a derivation that the current v2.3 machinery does not close.

V4.1 Υ-free per-galaxy re-run. Re-run the α fit without the fixed $\Upsilon_{3.6} = 0.5$ convention, allowing a per-galaxy Υ from Lelli+2016's bulge/disk decomposition. Test: does κ_ω survive as α^* when Υ is free, or does the best-fit α drift? Expected wall-clock: 1–2 weeks.

V4.2 2-parameter $M_{\rm eff}$ robustness. Fit the more general form $M_{\rm eff} = \alpha_\star M_\star + \alpha_{\rm gas} M_{\rm gas}$ (two free coefficients, not one) and check whether $\alpha_\star \neq 1$ at high significance would falsify the D-sector full-coupling claim for stars. A significant drift of $\alpha_\star$ from unity would be a genuine internal inconsistency.

V4.3 High-Σ subsample stability. Does α^* remain near κ_ω when restricted to the high-surface-brightness subsample (where Υ is most reliable), or does the match drift?

V4.4 Pointwise RAR discrimination. Given the v2.3 asymptotic relation $V_\infty^4 = GM_{\rm eff}c^2/(2\ell_\tau)$ with $\alpha = \kappa_\omega$, what does the pointwise RAR predict? Does it match SPARC's tight $g_{\rm obs}(g_{\rm bar})$ curve across ~ 5 decades? This is the sharpest MOND-vs-ΛCDM-vs-τ discriminator.

V4.5 Cosmology consistency: recombination-era baryon sector assignment. Resolve the apparent tension between the v2.3 trinary decomposition (which would assign recombination-era, fully-ionized baryons to the B-sector at coupling ≈ 0) and the v2.1 CMB prediction $\ell_1 = 220.6$ (which requires the full capacity-gradient $(1-\iota_\tau)/\iota_\tau^2 \approx 6.65\times$ baryon enhancement). This is flagged as a potential internal inconsistency in Physics Ledger Ch. 62.

V4.6 BBN consistency. Check that $Y_p = 20/81$ (V.T205) survives the trinary decomposition at $z \sim 10^9$, when baryons are fully ionized and plausibly B-sector.

V4.7 Cluster mass-budget consistency. With B-sector hot ICM at coupling ≈ 0, how does τ source the $\sim 10\times$ baryonic enhancement needed to account for cluster lensing (of which $\sim 80\%$ is dark matter in ΛCDM)? The capacity-gradient source must come from coherent instances; their density at cluster scale needs to be computed.

V4.8 Elliptical BTFR v4 test. ATLAS$^{\rm 3D}$ + SLUGGS ($N \sim 260$ galaxies) provides a cold-HI-free laboratory. The v2.3 prediction is $M_{\rm eff} \approx M_\star$ with no $f_{\rm gas}$ residual. This genuinely discriminates τ from both MOND (which predicts the BTFR with full M_b) and from ΛCDM (which requires dark halos of similar weight to ellipticals).

V4.9 First-principles derivation of sector structure. The D/ω/B coupling assignments are posited in v2.3 from the SPARC fit, not derived. v4 must derive them from the character structure of the ω-generator on $H_\partial[\omega]$, together with the $\omega \to B$ transition criterion (see paragraph 4.3 above).

Acknowledgements

The v3 scoping Bullet sketch (`bullet-cluster-capacity-gradient-sketch.tex`) anticipated the $\rho_\star$-vs-ρ_{gas} decomposition; the SPARC data provides the first galaxy-scale confirmation. McGaugh's R4 peer-review pass flagged the Ch. 63-to-Ch. 43 punt that forced this reflection. The v2.3.1 honesty revision owes its specific correction list to the Round-5 simulated panel (R1/R2/R3/R4 proxies) which identified the candidate-count miscount (R1+R3), the over-claimed uniqueness (R4+R3), the new-to-v2.3 sector-terminology flag (R3), the $\omega \to B$ transition open problem (R2+R4), the recombination-era baryon cosmology-consistency question (R2 Mack), and the $v4$ target list itself.

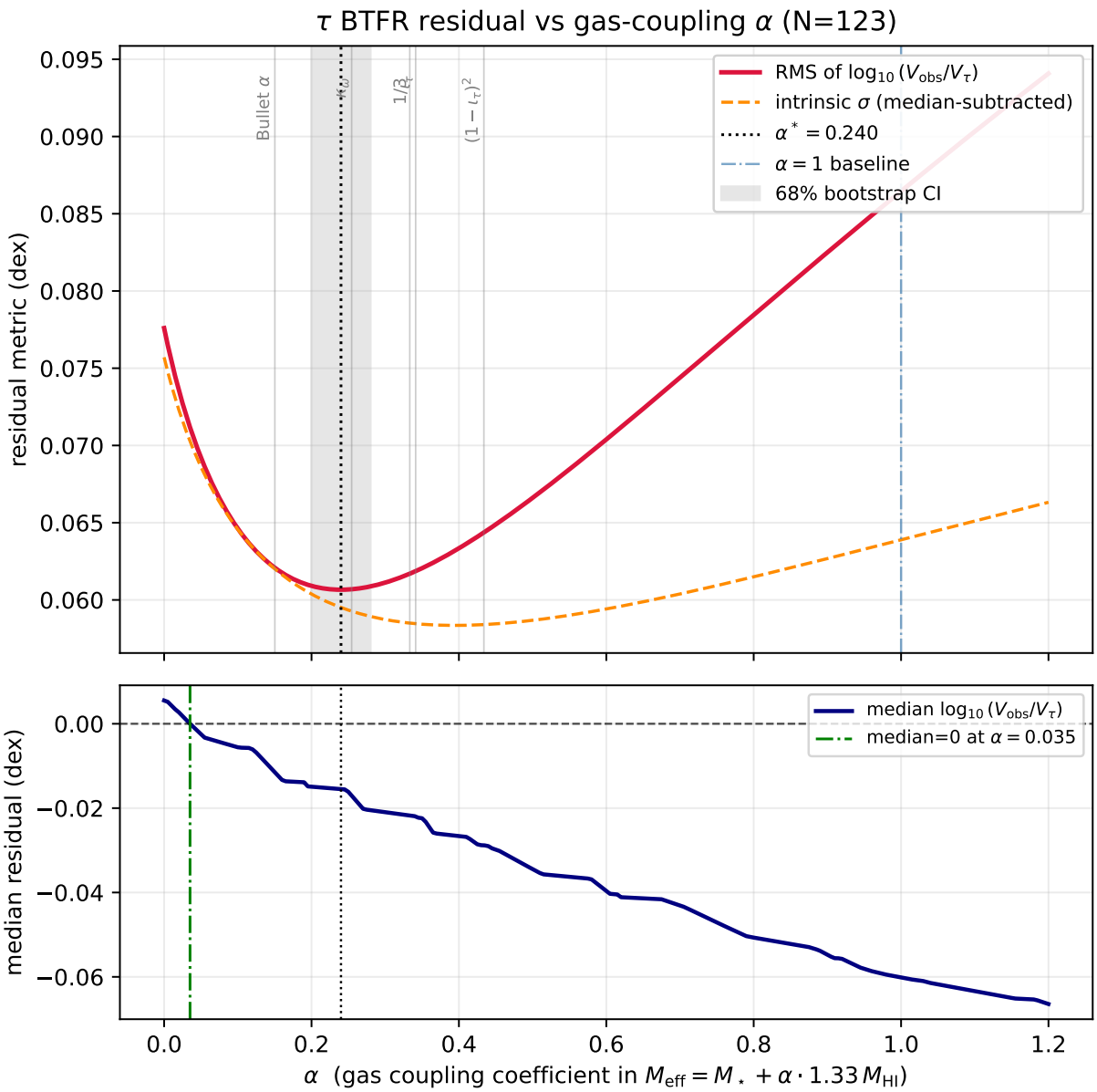

Figure 1: **Top.** Residual RMS (crimson) and intrinsic scatter (dashed orange) vs α. Minimum RMS at $\alpha^* = 0.240 \pm 0.040$ (dotted black, with shaded 68% bootstrap CI). Vertical gray lines mark τ-natural candidates. **Bottom.** Median residual crosses zero at $\alpha \approx 0.035$; at α^* a residual -0.015 dex (3.6% over-prediction) remains, 2.3σ from zero. The baseline $\alpha = 1$ is shown as a dash-dot blue line for reference.

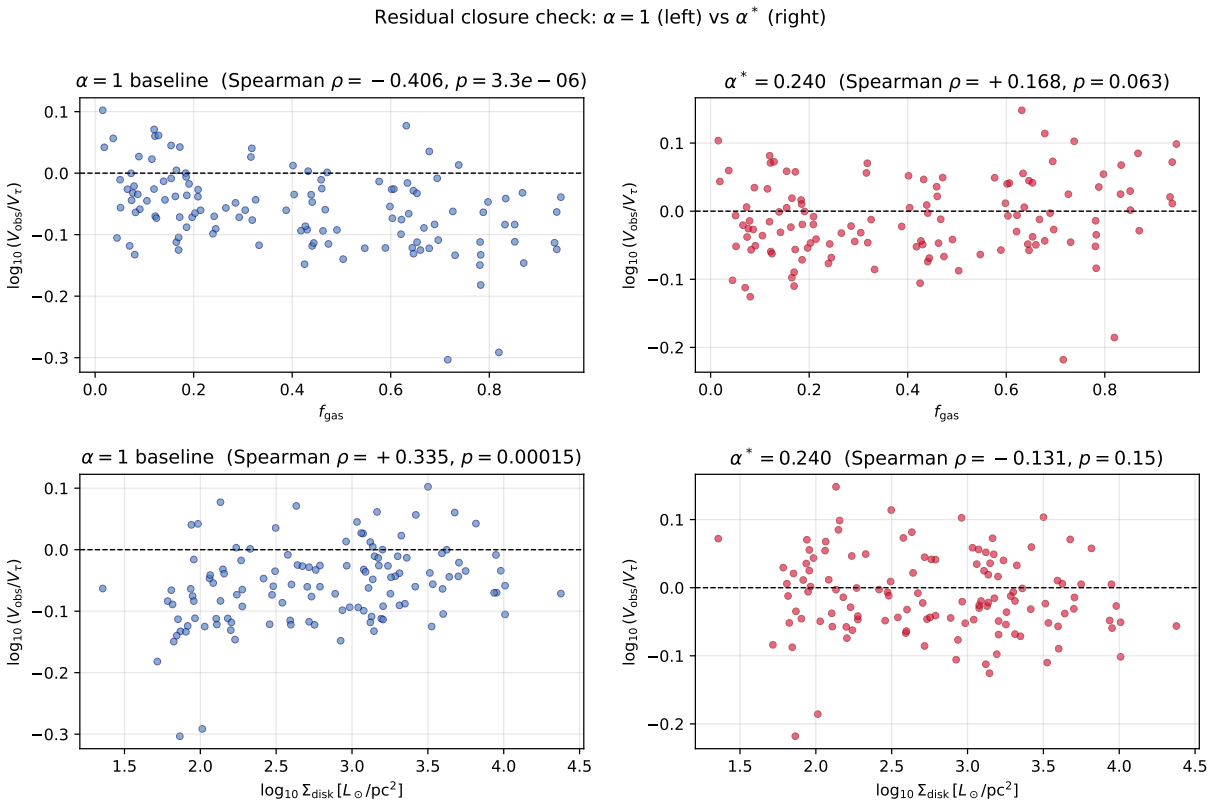

Figure 2: Residual correlations at $\alpha = 1$ baseline (left column) vs α^* (right column). Top row: $f_{\rm gas}$; Spearman ρ drops from -0.41 ($p = 3.3 \times 10^{-6}$) to $+0.17$ ($p = 0.06$). Bottom row: $\log \Sigma_{\rm disk}$; ρ drops from $+0.34$ ($p = 1.5 \times 10^{-4}$) to -0.13 ($p = 0.15$). Both residuals close at the same α^*.

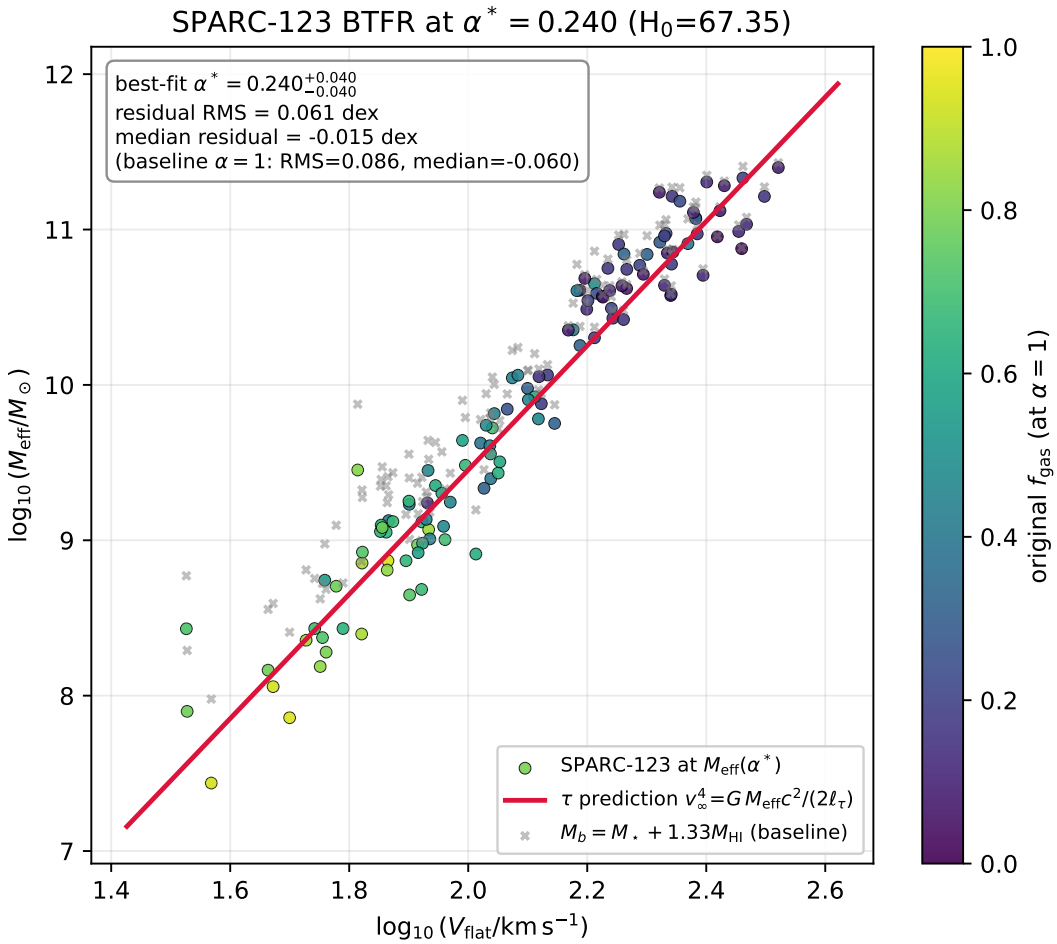

Figure 3: SPARC-123 BTFR at $\alpha^* = 0.240$: scatter of $\log M_{\rm eff}$ vs $\log V_{\rm flat}$ (circles, color-coded by original $f_{\rm gas}$) against the τ prediction line (crimson, slope 4, matched intercept). Grey $\times$'s show the v2.2 baseline $\log M_b$ for the same galaxies; the vertical displacement from circles is the gas-coupling correction.